

Demonstration of Joint Communication, Sensing & Monitoring with Unified Management on a 4-Core Multi-Core Fiber

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Abstract

We experimentally demonstrate a 4-core multi-core fiber SDM system, enabling simultaneous bi-directional coherent transmission, IM/DD transmission, integrated longitudinal power monitoring, distributed acoustic sensing, and unified control-plane operation across communication, monitoring, and sensing functions.

Review & Introduction

The increasing capacity demands of future optical networks have driven significant interest in space division multiplexing (SDM) based on multi-core optical fibers (MCFs) [1]. In particular, weakly coupled MCFs with inter-core crosstalk (ICXT) levels around -30 dB [2] enable the simultaneous transmission of heterogeneous services on separate cores within the same fiber. Beyond high-capacity communication, this creates the opportunity to joint integrate sensing and monitoring functionalities into a shared optical infrastructure. Such convergence is particularly relevant for future metro, access, and mobile transport networks, where efficient fiber utilization, operational awareness, and infrastructure supervision are becoming increasingly important.

Recent work has demonstrated the potential of SDM transmission [1], distributed sensing [3], and DSP-based monitoring techniques individually [4]. Distributed acoustic sensing (DAS) based on coherent Optical Time-Domain Reflectometry (C-OTDR) analyzes Rayleigh backscattered probing signals to detect vibrations and external disturbances affecting the fiber [5]. Longitudinal Power Monitoring (LPM) enables reconstruction of the optical power evolution along a fiber link, exploiting the interaction between chromatic dispersion and fiber nonlinearities without requiring dedicated monitoring hardware [6] within a coherent transmission system [7]. Integrating these functionalities with high-capacity transmission within a common SDM platform can provide enhanced operational visibility and infrastructure awareness while minimizing additional deployment complexity.

At the same time, single-fiber bidirectional (BiDi) architectures have gained relevance in the metro/access segments due to their efficient use of fiber. Combined with single-laser transceiver, these systems offer reduced hardware cost while maintaining the stringent synchronization and latency requirements of next-generation mobile net-

works. Conventionally, digital subcarrier multiplexing (DSCM) facilitated BiDi transmission by separating up- and down-stream channels with frequency interleaving, mitigating Rayleigh backscattering [8]. However, with power optimization, frequency interleaving might be no longer needed; and capacity can be doubled and performance improved even in the presence of impairments.

The simultaneous deployment of communication, sensing, and monitoring services within a shared MCF infrastructure introduces coexistence challenges related to interference, resource allocation, and system coordination. When joining the above mentioned technologies, the monitoring complexity of the resulting system increases, motivating the introduction of a dedicated management service, collecting and displaying system parameters in a unified manner. It also enables evaluation of potential interference between subsystems caused by ICXT.

In this demo, we present an unified management framework for joint SDM transmission, sensing, and monitoring system transmitted over a weakly coupled 4-core MCF. The setup simultaneously supports (i) 400G BiDi transmission using DSCM, (ii) intensity-modulation direct-detection (IM/DD) both for mobile fronthaul, (iii) DAS based on C-OTDR, and (iv) LPM within the same MCF infrastructure. The centralized management software platform interfaces with all subsystems and provides a unified real-time view of communication, sensing, and monitoring services, demonstrating the potential of integrated multi-service optical infrastructures for future intelligent optical networks.

Novelty

While joint sensing and communication SDM systems have been demonstrated in the past [9], to our knowledge, this demonstration will be the first SDM system to combine DAS, LPM, BiDi coherent DSCM and IM/DD, where each subsystem exclusively occupies one core of an MCF. Additionally, all systems are connected to a shared manage-

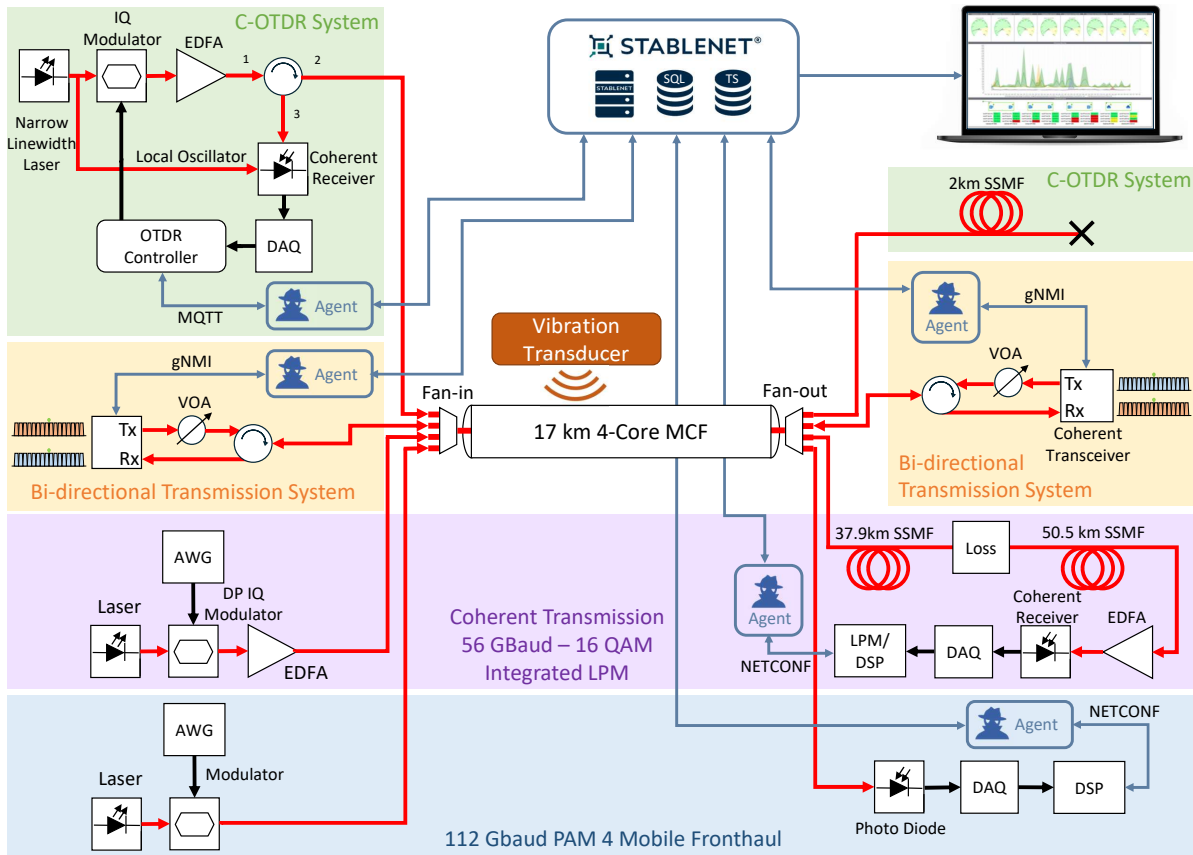


Fig. 1: Block diagram of the proposed demonstration setup.

ment layer (STABELNET®) and relay their monitoring and sensing parameters, providing centralized access to the entire system telemetry.

ECOC Relevance

The topic of joint sensing and communication has gained attention in recent years, as evidenced by its prominence at major conferences like ECOC. A common realization of such systems is SDM utilizing MCFs. With this proposal, we demonstrate a practical, joint sensing, communication and monitoring SDM system, reflecting near-future deployment scenarios. Compared to standard single-mode fibers (SSMFs), MCFs offer significantly higher spatial density, providing a more efficient solution for infrastructure scaling. Future infrastructure will likely involve heterogeneous integration, where MCFs are concatenated with legacy SSMFs and various optical technologies operating on a shared fiber plant, especially with SDM. These circumstances are emulated as part of the demonstration, with additional system monitoring provided by a dedicated software service.

Demo Implementation & Experimental Setup

Fig. 1 illustrates the proposed demo, which is located at Kiel University, and the attendees will access the demo management GUI via a remote connection. Here, crucial information like bit-error-ratios, intrusion events and fiber losses are displayed and updated in real time alongside a live

video stream of the physical setup. Additionally, attendees can remotely activate a vibration device to emulate an intrusion event.

The core component of the proposed demonstration is a 17 km long weakly-coupled 4-core MCF. While the majority of the fiber is kept on a spool, about 8 m of the fiber is spooled separately, where vibrations can be induced on the MCF. The following technologies will be used on the 4 different cores, as illustrated in Fig. 1.

Core 1 – DAS based on C-OTDR: For DAS, an ultra-narrow linewidth laser with $\lambda = 1550$ nm is split using a 3 dB splitter. The first branch is modulated using an IQ-Mach-Zehnder modulator with probing sequence bursts, which is then amplified using an EDFA and fed into the sensing core of the MCF via a circulator. The Rayleigh back-scattered light is then detected using a polarization diverse coherent receiver, with the second branch from the laser acting as the local oscillator, resulting in 4 electrical signals (I_X , Q_X , I_Y , Q_Y) which are finally digitized at a sampling rate of 500 MSa/s. The probing sequence bursts are modulated onto the probing branch at a sampling rate of 50 MSa/s, resulting in a symbol width of 10 samples per symbol. Within the DSP, the received signals for the polarization are matched filtered with the probing sequence burst. The DSP applied to the data is described in detail in [3]. Finally the sensing data is analyzed using machine-learning-

assisted DSP algorithms in order to detect and localize acoustic events. Events are emulated using an acoustic transducer towards the back end of the MCF. Detected events then raise an alarm within the management system, completing the intrusion detection pipeline.

Core 2 – BiDi transmission: A relevant scenario for DSCM in next-generation mobile networks is single-core single-wavelength full-duplex transmission, to better exploit available spectrum and reduce fiber deployment costs [8, 10, 11]. DSCM digitally generates, shapes, and combines multiple subcarriers onto a single optical carrier, enabling fine-granularity control of bit rate and bandwidth allocation [12, 13]. This represents a cost-effective, scalable, and software-defined optical transport for future metro-access networks [14].

In this demo, we transmit the full 16 subcarriers (400G transmission) from both DSCM transceivers over a single core of the MCF. The transmitter sides of the transceivers on both ends of the MCF are connected to the variable optical attenuators (VOA), which are used to control the transmit power. After that, the circulators are used to enable single core transmission. Compared to the frequency interleaving mode of operation between upstream and downstream, transmitting with the full-duplex allows doubling the capacity if the reflections can be fully mitigated. We will show different Tx power configurations to highlight the importance of Rayleigh backscattering and the power optimization. The transceivers are connected to the network management tools, StableNet©, for an overview and comprehensive network management.

Core 3 - Coherent data transmission with integrated LPM: Furthermore, the experimental setup illustrated in Fig. 1 carries a coherent optical transmission in the C-band. At the transmitter side, an AWG drives a dual-polarization IQ (DP-IQ) modulator encoding 16-QAM symbols at 56 Gbaud. The modulated signal is amplified by an EDFA and launched into core 3 of the MCF at a launch power of 10 dBm. To demonstrate the additional capabilities of LPM, core 3 is connected to an SSMF link totaling 88.4 km (37.9 km + 50.5 km), with a variable loss attenuator in between 2 SSMFs emulating a controlled power drop at the connection point. At the receiver side, the signal is detected by a coherent receiver. The DSP chain performs standard coherent processing, including dispersion compensation, synchronization, and carrier/phase recovery. LPM is then applied using the linear least squares approach [4], to estimate the power profile for each fiber segment. The extracted power profile is fed to an agent, which leverages this information to enable autonomous monitoring of the transmission link

using StableNet©.

Core 4 - IM/DD transmission: The fourth core of the MCF is used for an IM/DD transmission system targeting mobile fronthaul. Data is transmitted at a rate of 112 Gbaud using PAM 4 intensity modulation in the O-band. On the receiver side, the data is demodulated using a photo diode and digitized [15]. The bit-error-rate is calculated and provided to the management layer, where it is stored and displayed as time-series data.

Management layer

All components and schemes are running in parallel providing different interfaces for integration with other tools. In this demo, all components are integrated in a unified network management solution to link communication, sensing, and monitoring data together and gain a comprehensive operational overview. StableNet©, Infosim's software solution for automated network and service management, leverages a distributed server-agent architecture. At its core is the server, which connects to time-series and relational databases where optical telemetry data, LPM power profiles, and DAS logs are consolidated and persisted. Devices are connected via lightweight agents that receive raw data, perform edge pre-processing (filtering, aggregation, normalization), and stream metrics to the central server at configurable intervals. The agent already provides a wide range of interfaces to connect to devices by default, e.g. via snmp for legacy endpoints, but is extended with modern, open programmable interfaces using gNMI for high-bandwidth telemetry and NETCONF/YANG for structured configuration. To support the operational demands of the multi-core SDM infrastructure, StableNet's© control plane maps optical-layer telemetry to service-aware models, allowing cross-domain correlation and automated fault handling. LPM-derived power profiles and DAS data are assigned to the StableNet's© data structure, allowing agents to trigger threshold-based alarms or initiate other workflows.

Conclusion

We propose a demo for joint communication, sensing, and monitoring of a SDM system based on a 4-core MCF, with centralized access to subsystem and network parameters through the StableNet© management platform. The setup combines communication, sensing, and monitoring services within a shared fiber infrastructure, enabling unified system supervision and improved operational awareness. Potential intrusion attempts can be emulated and detected while simultaneous data transmission and system health monitoring remain active. Furthermore, ICXT-related interference effects can be evaluated and quantified by operating different combinations of the four subsystems simultaneously.

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